Institute of Technology of Cambodia

Departement of Applied Mathematics and Statistic

Topic: Web Scraping & Data Analysis

Reddit Data Science

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INTRODUCTION

What is Reddit ?

>> Reddit is a large online platform where users share posts, links, images, and questions

>>Content is organized into topic-based communities called subreddits

>>Users interact through upvotes, downvotes, and comments, which determine post visibility

>>With its large and active user base, Reddit is a valuable source of real-world data for analysis

OBJECTIVES

1.How does the number of posts in the r/datascience subreddit change over time?

2.How do users engage with posts in terms of scores and comment counts?

3.How are comment counts distributed, and do they show skewness or outliers?

4.Is there a relationship between the number of upvotes (score) and the number of comments on a post?

5. How does user engagement vary across different post flairs?

DATA COLLECTION (WEB SCRAPING)

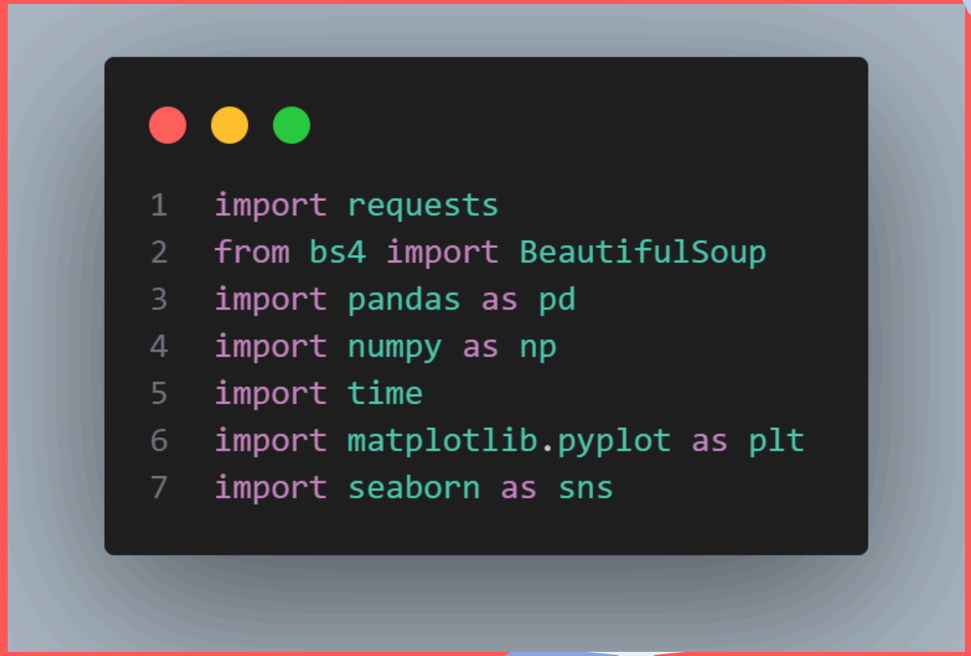
Website : https://old.reddit.com/r/datascience

The web scraping and data collection process was implemented using the Python programming language. The following libraries were used:

> bs4

>requests

The scraping process involved sending HTTP requests to multiple pages of the r/datascience subreddit and parsing the HTML content using BeautifulSoup



```
import requests
from bs4 import BeautifulSoup
import pandas as pd
import numpy as np
import time
import matplotlib.pyplot as plt
import seaborn as sns
```

Dataset Description

	Title	Score	Comments	Author	Time	Flair
0	Weekly Entering & Transitioning - Thread 26 Ja...	16	10	AutoModerator	2026-01-26T05:01:28+00:00	No Flair
1	[Official] 2025 End of Year Salary Sharing thread	115	141	Omega037	2026-01-01T04:38:20+00:00	No Flair
2	Google Maps query for whole state	0	3	big_data_mike	2026-01-29T01:38:30+00:00	Projects
3	What is the Best Minecraft Server Hosting?	0	0	GODLIKE-host	None	No Flair
4	Resource: Awesome Marketing Science - A curate...	31	1	Dizzy-Midnight-6929	2026-01-27T21:37:47+00:00	Education
...	...	...	...	...	...	...
1078	Weekly Entering & Transitioning - Thread 03 Fe...	11	39	AutoModerator	2025-02-03T05:01:44+00:00	No Flair
1079	[AI Tools] What AI Tools do you use as a copil...	70	38	limedove	2025-02-02T10:16:24+00:00	Tools
1080	any one here built a recommender system before...	4	11	Emotional-Rhubarb725	2025-02-02T19:59:00+00:00	Projects
1081	Use LLMs like scikit-learn	129	5	No_Information6299	2025-02-01T16:46:39+00:00	Projects
1082	deepseek.com is down constantly. Alternatives ...	0	14	mehul_gupta1997	2025-02-02T14:55:58+00:00	AI

After scrapping data collection , It has 1083 row and 6 columns.

Data Types:

- Title: The textual headline of the Reddit post, representing the main topic of discussion.
- Score: The net score of the post, calculated as upvotes minus downvotes, indicating community approval.
- Comments: The total number of comments received by the post, reflecting user engagement.
- Author: The username of the individual who submitted the post.
- Post_Time: The date and time when the post was published, used for time-based analysis.
- Flair: The category or tag assigned to the post, such as Discussion, Question, or Project.

Data Preprocessing

```
Data columns (total 6 columns):
 #   Column    Non-Null Count  Dtype
---  ------    --------------  -----
 0   Title     1083 non-null   object
 1   Score     1083 non-null   int64
 2   Comments  1083 non-null   int64
 3   Author    1083 non-null   object
 4   Time      984 non-null    object
 5   Flair     1083 non-null   object
dtypes: int64(2), object(4)
memory usage: 50.9+ KB
```

```
df.isnull().sum()
```

```
Title        0
Score        0
Comments     0
Author       0
Time        99
Flair        0
dtype: int64
```

We found that 99 missing values of Time.

>>>Drop missing values in critical columns.

```
1  initial_rows = len(df)
2  df.dropna(subset=["Title", "Score", "Comments", "Time"], inplace=True)
3  cleaned_rows = len(df)
4  print(f"Rows removed due to missing values: {initial_rows - cleaned_rows}")
```

Out put: Rows removed due to missing values: 99

0 Find Duplicated and Remove

```python
duplicates = df.duplicated(subset=["Title", "Author", "Time"]).sum()
df.drop_duplicates(subset=["Title", "Author", "Time"], inplace=True)
print(f"Duplicate rows removed: {duplicates}")
```

Since dataset has 1083 rows, have missing value 99 row , and duplicate 0

After remove missing value 99 rows

SO,Final Dataset Size: 984 rows

Out put: Duplicate rows removed: 0

>>>Convert numerical columns to integer

```python
df["Score"] = pd.to_numeric(df["Score"], errors="coerce").astype(int)
df["Comments"] = pd.to_numeric(df["Comments"], errors="coerce").astype(int)

```

```python
df['engagement'] = df['Score'] + df['Comments']
```

A new feature called Engagement was created by combining the post score and the number of comments

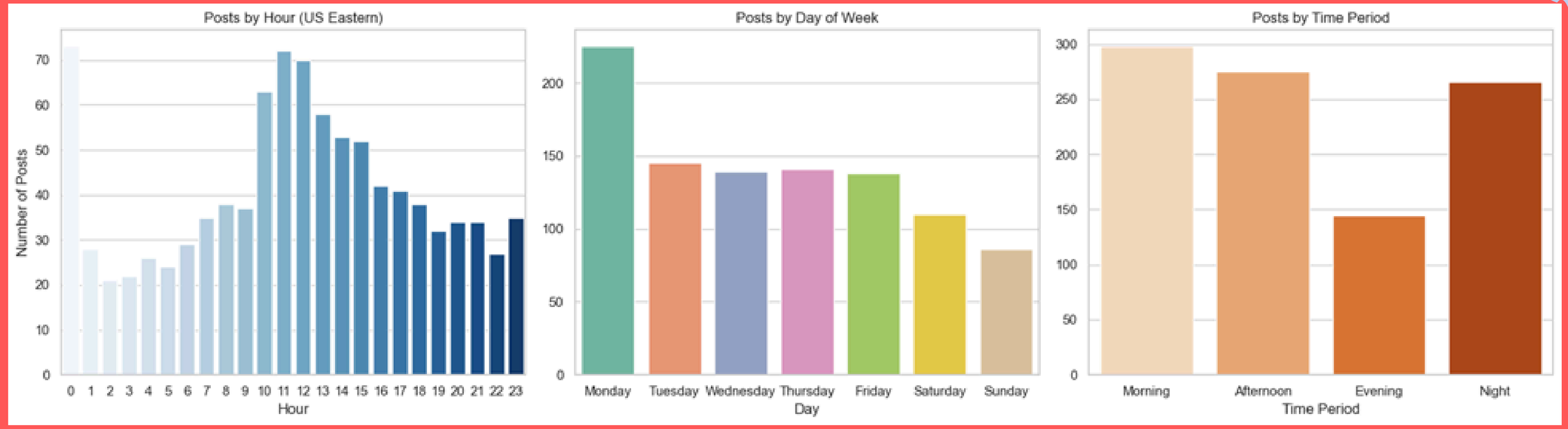
Descriptive Statistic

	Score	Comments	Title_Length	Comments_per_Score
count	984.000000	984.000000	984.000000	984.000000
mean	103.317073	34.264228	60.077236	2.235517
std	256.030559	45.078560	30.687811	5.667300
min	0.000000	0.000000	10.000000	0.000000
25%	7.000000	9.000000	39.000000	0.302013
50%	26.000000	19.000000	56.000000	0.616922
75%	86.000000	41.000000	72.000000	1.314103
max	2760.000000	433.000000	245.000000	74.000000

Insight Analysis

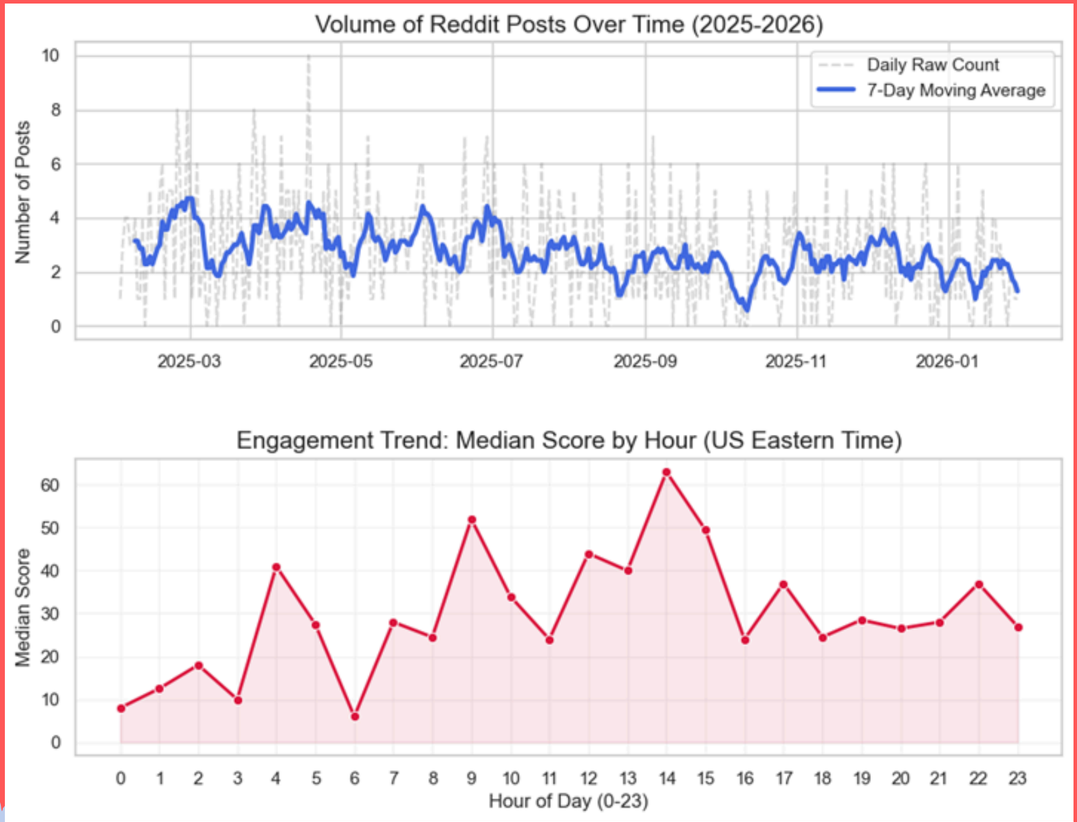


Posting Activity Over Time



- *Posting activity is not evenly distributed across time*
- *More posts are created during specific hours of the day*
- *Weekdays show higher posting activity compared to weekends*
- *Activity fluctuates over time, reflecting changing community engagement*

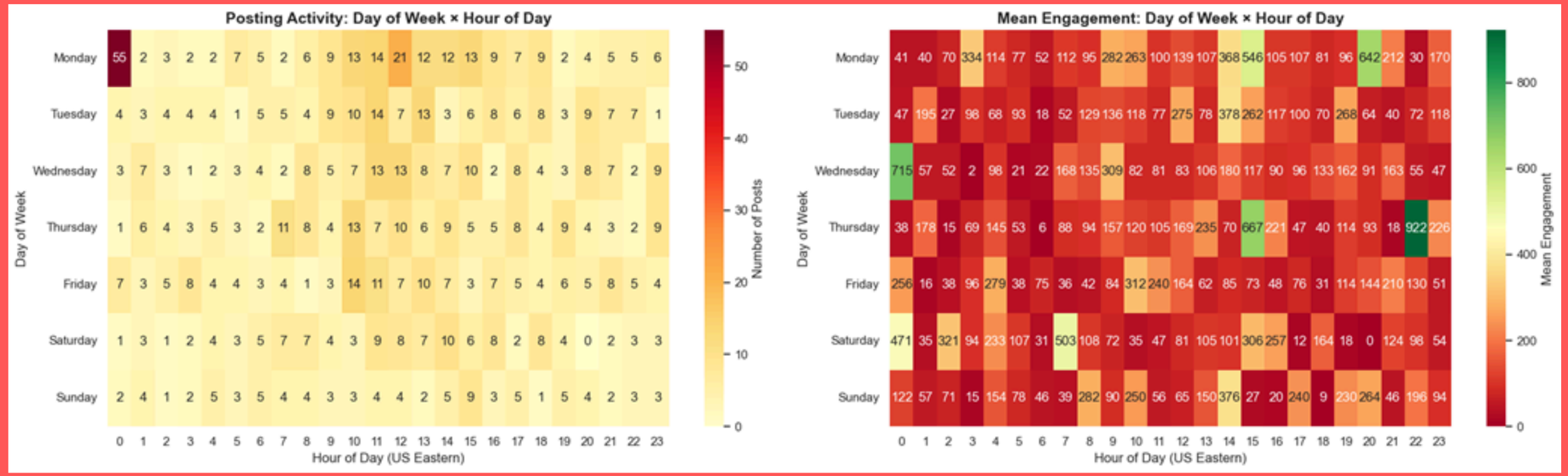
Posts Per Day with 7-Day Rolling Average and Engagement by Hour



- Posting activity fluctuates across time rather than remaining constant
- Short-term daily variation is visible, while the 7-day rolling average highlights broader participation trends
- Periods of increased and decreased activity suggest dynamic community engagement

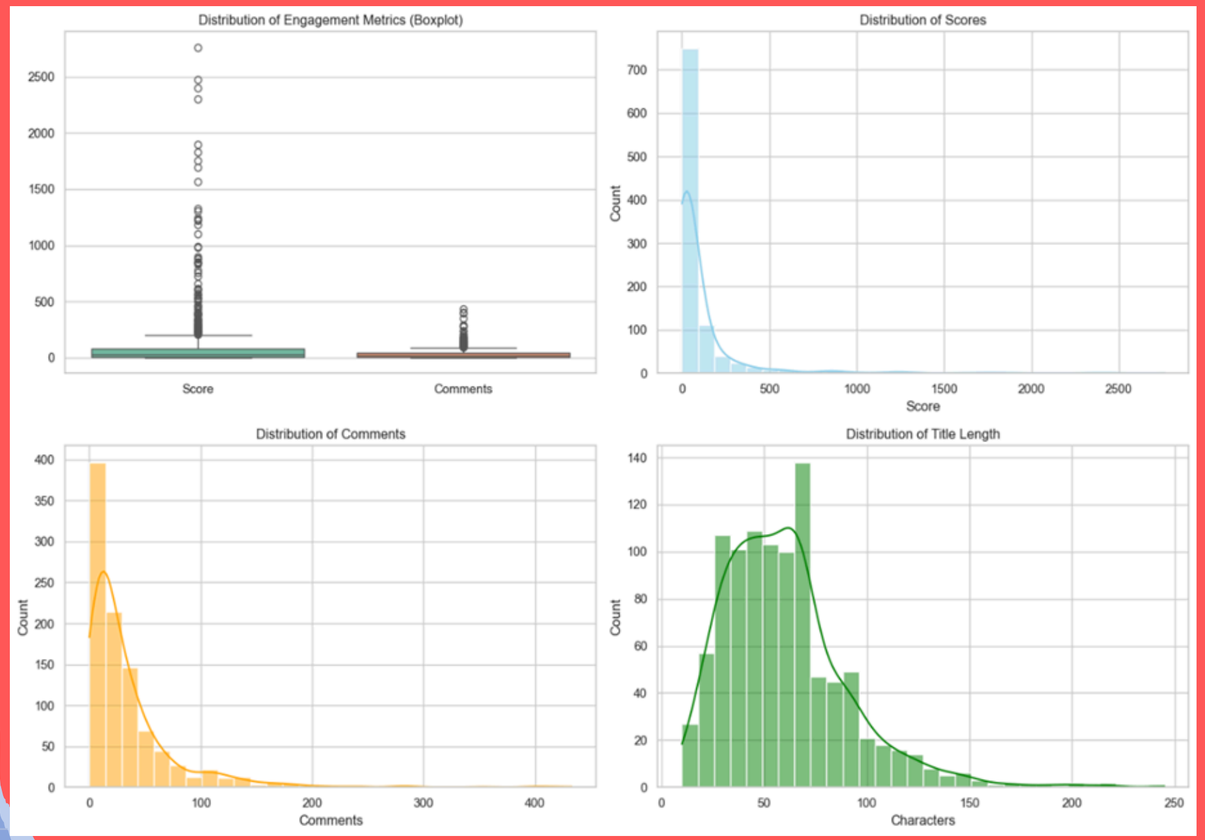
- Engagement levels vary across different hours of the day
- Higher interaction is observed during certain daytime and evening hours
- Late-night and early-morning periods show lower average engagement

Day of Week × Hour of Day Heatmaps



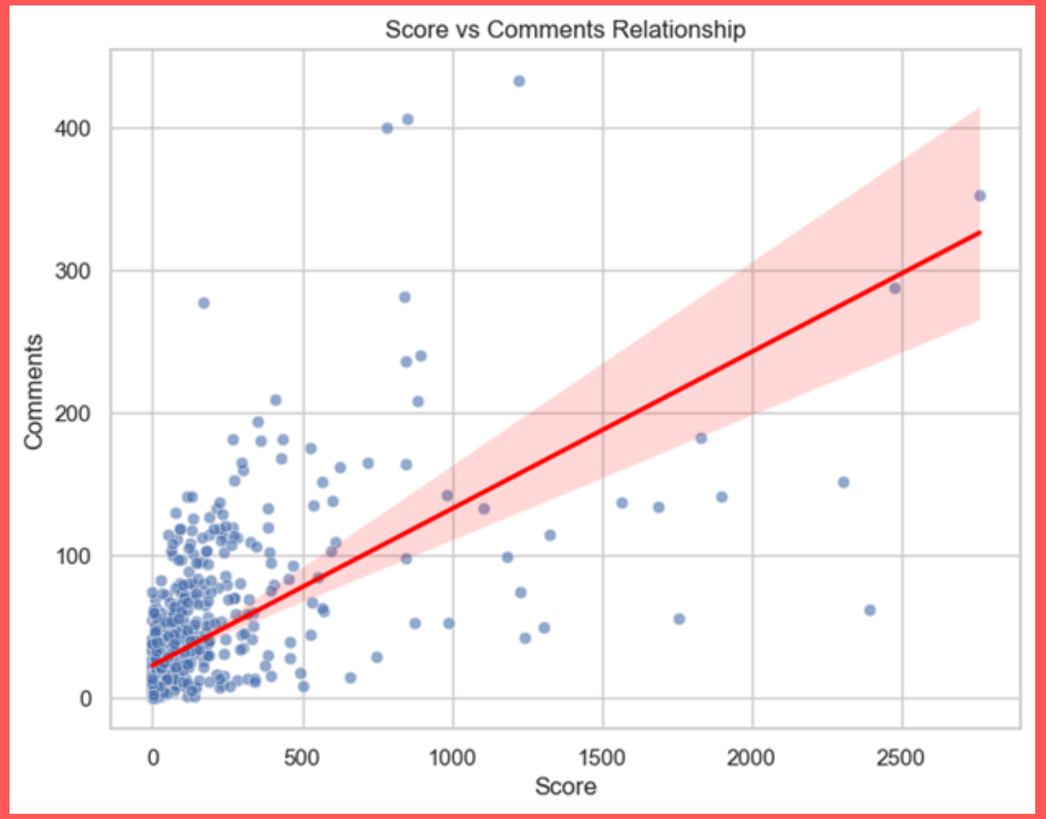
- *Posting activity is higher on weekdays than weekends*
- *Activity peaks during late morning and early afternoon*
- *Engagement varies across time and does not always match posting volume*
- *Some low-activity periods still show high engagement*

Engagement Distributions



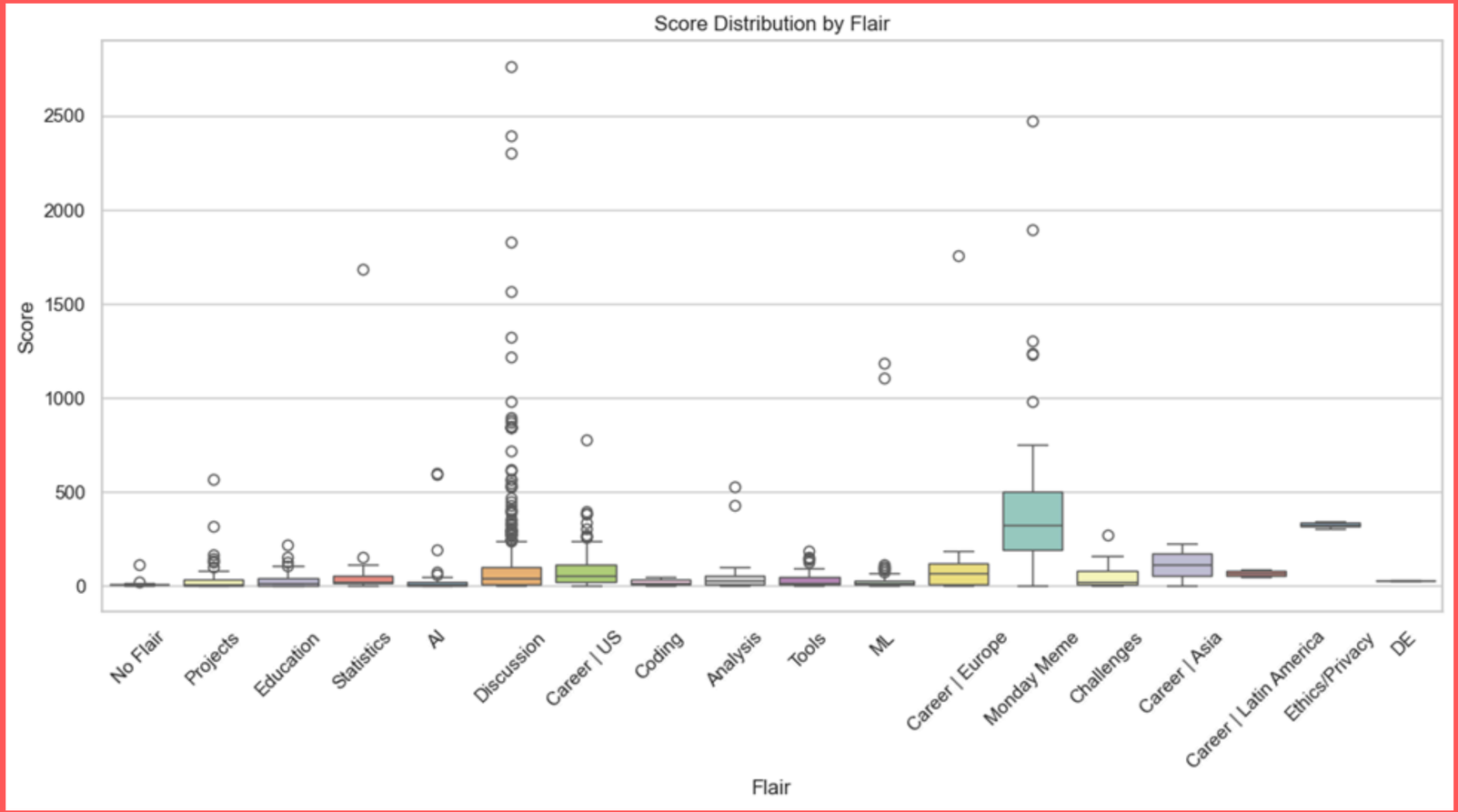
- *High concentration of data at lower values with extreme upper outliers. This confirms r/datascience follows a "Winner-Take-All" dynamic where a few posts capture the majority of community attention.*
- *Data is heavily right-skewed. The Mean Score (103) is nearly 4x higher than the Median (26), proving that "average" engagement is driven by rare viral events.*
- *Average length is 60 characters. Most successful titles are concise and descriptive; excessively long or very short titles correlate with lower engagement "floors."*

Relationship Between Score and Comments



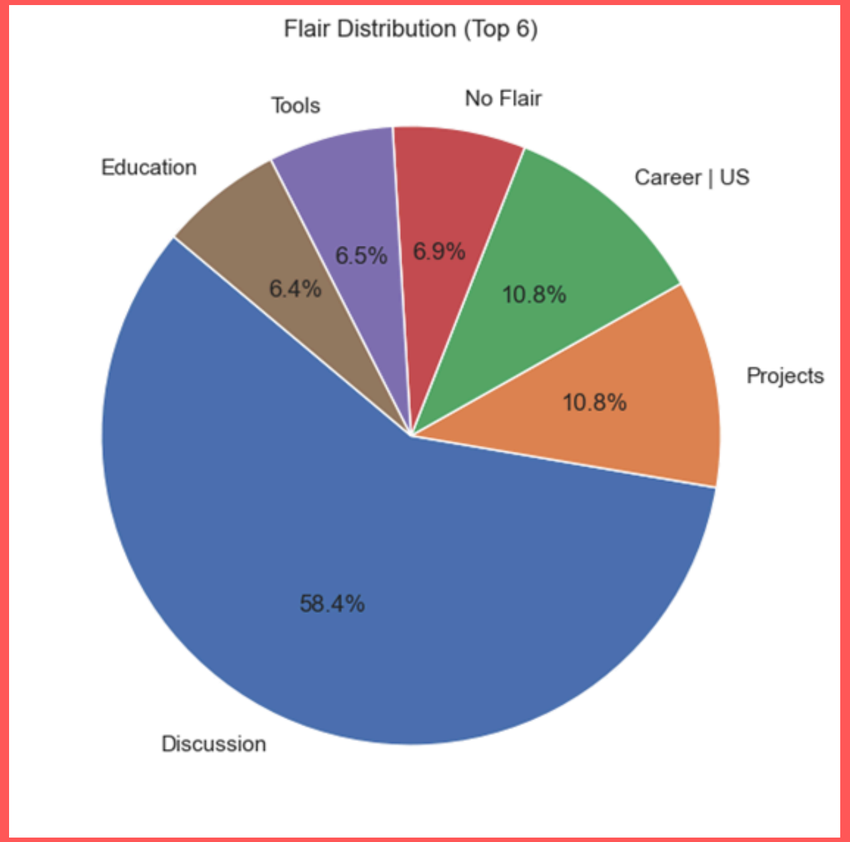
- Positive relationship
- Posts with higher scores tend to receive more comments

Engagement by Flair



- ***Discussion and career-related flairs show higher engagement***
- ***Community prefers practical and discussion-based content***

Top 5 Flairs + Others



Use it to show that "Discussion" and "Career" make up the majority of the community's volume.

Key Findings

- Engagement is highly right-skewed, with a small number of posts receiving very high interaction
- Posting activity and engagement vary by time of day and day of week
- Score and comment count show a positive association
- Discussion and career-related flairs tend to attract higher engagement

Limitations of the Study

- Time Window: Dataset is a snapshot and may not reflect long-term seasonal trends.
- Metadata Focus: Analysis excludes the full text of post bodies and comments.
- Methodology: Web scraping has limitations in data completeness compared to the official API.

CONCLUSION

This project successfully applied web scraping and exploratory data analysis to examine posting behavior and engagement patterns in the r/datascience subreddit. The analysis revealed clear temporal and categorical patterns in posting activity and engagement, including highly skewed interaction distributions and variation across post flairs and time periods. As an exploratory study, the findings describe observed patterns rather than causal relationships, providing an overview of how engagement is distributed within the community.

FUTURE WORK AND IMPROVEMENTS

NATURAL LANGUAGE PROCESSING (NLP)

INCORPORATE SENTIMENT ANALYSIS AND KEYWORD EXTRACTION FROM POST BODIES AND COMMENT SECTIONS TO DECODE WHY CERTAIN TOPICS GO VIRAL.

API INTEGRATION

SHIFT FROM WEB SCRAPING TO THE OFFICIAL REDDIT API TO IMPROVE DATA COMPLETENESS AND INCLUDE LONG-TERM HISTORICAL TRENDS.

PREDICTIVE ANALYTICS

Develop machine learning models to predict a post's potential score based on title length, flair selection, and submission time.

THANK YOU